

Total No. of Questions : 4]

SEAT No. :

PC60

[6360]-62

[Total No. of Pages :2

T.E. (Information Technology) (Insem)

INTERNET OF THINGS

(2019 Pattern) (Semester- I) (314445D) (Elective - I)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2 and Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

UNIT - 1

- Q1) a) Explain the Framework or Architecture of IoT. [5]
- b) Compare IoT and M2M with different parameters. [5]
- c) Write a short note on IoT Enabled Technologies - Wireless Sensor Networks. [5]

OR

- Q2) a) Explain the IoT Communication APIs. [5]
- b) Differentiate between Software defined Networking (SDN) and Network Function Virtualization (NFV). [5]
- c) Write a short note on IoT Enabled Technologies - Cloud Computing. [5]

UNIT - 2

- Q3) a) Define Actuators and explain types of actuators with examples. [7]
- b) Draw and explain interfacing of Buzzer with ARDIUNO with a program to ring the buzzer. [8]

OR

P.T.O.

Q4) a) Draw and explain interfacing of LED with ARDIUNO with a program to blink it. [7]

b) Write a short note on: [8]

i) AC Power devices

ii) Servo motor



Total No. of Questions : 4]

SEAT No. :

P9032

[Total No. of Pages : 1

Oct-22/TE/Insem - 564
T.E. (Information Technology)
INTERNET OF THINGS
(2019 Pattern) (Semester - I) (Elective - I) (314445 D)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Define IoT and explain physical design of IoT. [7]
b) Explain IoT world Forum (IoTWF) standardized seven-layer IoT architectural reference model. [8]

OR

- Q2)** a) Explain enabling technologies in IoT. [7]
b) Write a short note on (Any Two): [8]
i) Software defined Networking
ii) Network function virtualization
iii) Wireless Sensor Networks

- Q3)** a) Define Sensor and explain types of sensors with examples. [7]
b) Write a short note on (Any Two): [8]
i) 12C bus protocol
ii) CAN bus protocol
iii) UART
iv) USRT

OR

- Q4)** a) Draw and explain interfacing of LED with ARDIUNO with program to blink it. [7]
b) Draw and explain interfacing of DC motor with ARDIUNO with program of speed control. [8]

***** END *****

Total No. of Questions : 4]

SEAT No. :

P5061

[Total No. of Pages : 1

[6187]-464

T.E. (Information Technology)(Insem)
INTERNET OF THINGS
(2019 Pattern) (Semester-I) (Elective-I) (314445 D)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Solve Q.1 or Q.2, Q.3 or Q.4.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data if necessary*

- Q1)** a) Define IOT and explain physical design of IOT. [7]
b) Explain IOT World Forum (IoTWF) standardized seven-layer IoT architectural reference model. [8]

OR

- Q2)** a) Explain enabling technologies in IOT. [7]
b) Write a short note on :
Software defined Networking
Network function virtualization [8]

- Q3)** a) Define sensor and explain types of sensors with examples. [7]
b) Draw and explain interfacing of DC motor with ARDIUNO with program of speed control. [8]

OR

- Q4)** a) Draw and explain interfacing of LED with ARDIUNO with program to blink it. [7]
b) Write a short note on :
i) I2C bus protocol
ii) CAN bus protocol [8]

